# VALACICLOVIR

Fewer daily doses

Renal dose adjustment

**CHEMICAL NAME** L-valyl ester of aciclovir: 2-[(2-amino-6-oxo-1,6-dihydro-9H-purin-9-yl)methoxy]ethyl L-valinate (hydrochloride)

**CLASS** Nucleoside analogue antiviral (prodrug of aciclovir)

**TRADE NAMES** Valtrex · Zelitrex · Valcivir

## THERAPEUTIC USES / INDICATIONS

- Herpes zoster (shingles).
- Genital herpes — treatment of episodes and suppression; reduces transmission.
- Cold sores (herpes labialis).
- CMV prophylaxis after renal transplant.

## PHARMACODYNAMICS

- Converted to aciclovir → phosphorylated by viral thymidine kinase → inhibits **viral DNA polymerase**.
- Active against HSV-1, HSV-2, VZV; some CMV activity at high doses.

## ADMINISTRATION

- **PO** with or without food.
- Start treatment as early as possible (within 72 h for shingles).
- Maintain adequate hydration.

## SIDE & ADVERSE EFFECTS

### SIDE EFFECTS (COMMON)

- Headache, nausea.
- Abdominal pain, dizziness.
- Rash.

### ADVERSE EFFECTS (SERIOUS)

- Acute kidney injury.
- Neurotoxicity (confusion, hallucinations) in elderly/renal impairment.
- Thrombotic microangiopathy (very high doses in immunocompromised).

## DOSAGES

- **Shingles:** 1 g three times daily for 7 days.
- **Genital herpes first episode:** 1 g twice daily for 7–10 days; recurrence 500 mg twice daily for 3 days.
- **Suppression:** 500 mg once daily.
- **Cold sores:** 2 g twice daily for 1 day.
- Reduce dose if CrCl <50 mL/min.

## CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity to valaciclovir or aciclovir.
- **Caution:** renal impairment, elderly, dehydration.

### KEY INTERACTIONS

- **Nephrotoxic drugs:** ↑ renal risk.
- **Cimetidine, probenecid:** ↑ aciclovir levels.
- **Mycophenolate:** ↑ levels.

## PHARMACOKINETICS

- **Peak:** 1.5 h. Rapidly converted to aciclovir; bioavailability of aciclovir ~55% (3–5× oral aciclovir).
- Excreted as aciclovir in urine. **t½:** 2.5–3 h.

## NURSING MANAGEMENT

- Check renal function and adjust dose.
- Encourage fluids.
- Monitor mental status in elderly.
- Assess lesions and pain (post-herpetic neuralgia).
- Infection-control advice.

## PATIENT EDUCATION

- Start at the first tingle or rash.
- Drink plenty of water.
- Use condoms; avoid sex during outbreaks.
- Report confusion or reduced urine.

**KEY POINTS**

- Prodrug of aciclovir; better absorption.
- Fewer doses than aciclovir.
- Shingles 1 g TDS × 7 days.
- Adjust in renal impairment.